Digilians at ImageEval 2026 Shared Tasks: Modular Speech Parsing and Joint Multimodal Reasoning for Arabic Spoken VQA

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Abstract

We describe the Digilians submission to Task 1a (Spoken Visual Question Answering, MSA track) of the ImageEval 2026 shared task (Abdaljalil et al., 2026). Our system splits the problem into independently cacheable stages: speech recognition, schema-constrained transcript parsing that recovers a question and three candidate answers from continuous Arabic speech, a conditional repair mechanism that escalates to a larger ASR model only for semantically degenerate parses, and a joint multiple-choice visual reasoning stage that evaluates all three candidates against the image simultaneously rather than scoring them independently. On the labeled development split (500 records, synthetic voice-cloned audio), our production configuration reaches 79.0% accuracy against a 39.8% official baseline, with nearly the entire gain over an earlier configuration coming from a single change to the answer-selection model. On the blind final test split (1,000 records, human-recorded audio), the same frozen configuration scores 65.6% accuracy / 57.4% balanced accuracy / 50.4% macro F1, a substantial drop we attribute to the dev-to-test recording-condition shift and report as our most important finding. We also surface a caching-related reproducibility issue uncovered during development and discuss what both findings imply for pipelines built the same way.

1 Introduction

Spoken Visual Question Answering (SVQA) extends conventional VQA by removing the assumption that the question arrives as text. Task 1a of the ImageEval 2026 shared task (Abdaljalil et al., 2026) instantiates this for Modern Standard Arabic (MSA): each record pairs an image with a single spoken clip containing both the question and its three candidate answers, so the system implements $f(I, A) \rightarrow y$, $y \in \{0, 1, 2\}$, where I is the image and A the spoken audio. No textual channel for the question exists anywhere in the input: everything the system knows about what is being asked must first be recovered from speech.

This compounds three difficulties. MSA transcription is itself hard: the organizers' baseline scores 39.8% on MSA against 66.4% on the parallel English track for an identical task design. The audio never announces option boundaries ("Option One"), so recovering that structure is an extraction problem distinct from transcription accuracy. Finally, once a clean question and three options exist, the remaining problem is genuine multimodal reasoning — and multimodal models are known to answer differently when a question is spoken rather than read (Mousi et al., 2026a), and to produce confident but counterfactual answers in multilingual visual settings (Mousi et al., 2026b).

The official baseline treats this as a single end-to-end call: one vision-language model receives the audio and image and predicts directly, with a regular expression extracting the index from free text. This hides every intermediate failure inside one opaque call, so error analysis cannot tell whether a mistake came from hearing, segmentation, or reasoning. We instead decompose the task into independently testable stages with well-defined in-

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put/output contracts, and constrain every machine-read output with JSON schemas. A manual inspection of Task 1a records (Appendix A) motivated each stage; most consequentially, speakers rarely mark option boundaries, so continuous speech must be segmented before it can be reasoned over.

Our contributions are: (i) a modular SVQA pipeline whose perception, language-understanding, and reasoning stages are independently cacheable (§2); (ii) schema-constrained transcript parsing plus a conditional repair mechanism that detects semantically degenerate but schema-valid parses (§2.2–§2.3); (iii) joint multiple-choice visual reasoning over all three candidates in one prompt (§2.5); and (iv) a single-variable experimental protocol isolating each design choice (§4), including a dev-to-test drop we can trace to a likely cause rather than only observe (§4.3).

Our ASR stage builds on Whisper (Radford et al., 2023); parsing and selection call Qwen-family vision-language models (Bai et al., 2023, 2025), served locally rather than through a hosted API. We treat the official Task 1a baseline as our primary comparison, since it is the only other system evaluated on this exact task formulation at the time of writing.

2 System Description

Figure 1 summarizes the pipeline. Every arrow between layers is also a cache boundary (§2.6).

2.1 Perception: Speech Recognition

The primary ASR pass runs Whisper (medium) locally via `faster-whisper` on CPU with int8 quantization on every record; records flagged by repair (§2.3) are re-transcribed with Whisper large-v3 on GPU, keeping average cost low while reserving a stronger fallback for the hardest cases. Two additional backends (QCRI Fanar Aura-STT, OpenAI `gpt-4o-transcribe`) are implemented and unit-tested against mocked traffic but never run live; all results use Whisper exclusively. Every ASR result, including a failed transcript, is cached immediately, so a transient failure is not silently retried.

2.2 Language Understanding: Parsing

A text-only call to a local vision-language model (Qwen2.5-VL, used purely as a text model) turns the raw transcript into a structured object, constrained by JSON schema to four fields: `question`, `option_0`, `option_1`, `option_2`. If the upstream transcript failed or is empty, this stage short-circuits without writing to cache, so fixing an ASR problem later triggers an automatic retry with no manual invalidation. The parsing prompt went through two revisions; §4.2 discusses the issue this exposed.

2.3 Reliability: Conditional Repair

A parse can satisfy the schema while still being unusable. Manual error analysis surfaced recurring, schema-valid but degenerate patterns: empty or very short options; duplicated, overlapping, or subsumed options; boilerplate leakage (a parsed "option" that is really transcript scaffolding, e.g. *al-khiyārāt hiya*, "the options are"); fabricated placeholders ungrounded in the transcript; and, on noisy recordings, ASR repetition loops. These motivated a deterministic, rule-based validator over a learned checker, so repair triggers stay auditable and reproducible.

Flagged parses trigger a three-step escalation, stopping at the first success: (1) a **stricter reparse** of the same transcript; (2) **re-transcription with Whisper large-v3** plus reparsing; (3) **proceed unrepaired** rather than block. Each step has its own cache namespace, letting error analysis distinguish "repair fixed it" from "attempted and failed" from "never triggered" for the same record.

2.4 Few-Shot Retrieval (Off by Default)

An optional subsystem classifies the query image into a fixed taxonomy (9 categories / 31 subcategories) via a schema-constrained call, then shows the selector up to k category-matched exemplars from `train`, hydrated into full (question, options, gold answer) triples. We evaluate it in §4; it remains off in production.

2.5 Reasoning: Joint Visual Selection

The selector receives the image and all three candidates in one multimodal prompt and returns `answer_index`, constrained by JSON schema to $\{0, 1, 2\}$, rather than scoring each option independently. When candidates are semantically close (e.g. *dog* / *wolf* / *fox*), independent scoring prevents the model from comparing competing hypotheses directly, whereas joint presentation lets it reason comparatively. Working from an already-validated question and option set also means this stage never splits effort across speech understanding, segmentation, and vision at once.

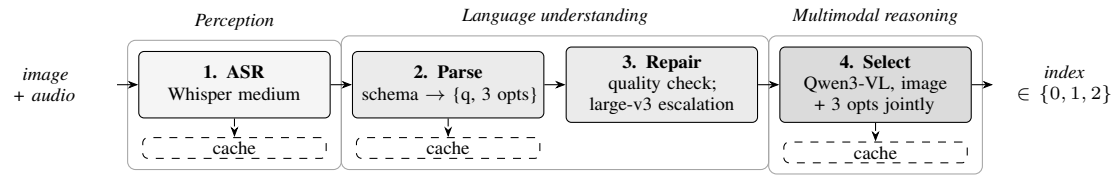


Figure 1: The four-stage pipeline. Only stage 4 sees the image; stages 1–3 recover a structured question and three candidates from continuous speech. Each stage writes to its own on-disk cache (dashed), so one component can be swapped and re-evaluated without recomputing the others.

The production selector is Qwen3-VL (8B), served locally via Ollama; an earlier configuration used Qwen2.5-VL (7B) with every other stage held identical. The effect of this single swap is, by a wide margin, our largest gain.

2.6 Engineering Infrastructure

Because every stage owns an independent cache namespace, changing only the selector reuses cached transcripts and parses, and an interrupted run resumes from its last completed record. Every reported metric comes from the organizers' own vendored, unmodified scorer, not a custom script. This made the six-plus single-variable iterations in §4 feasible; its one demonstrated failure mode, an under-specified cache key silently mixing outputs from two pipeline versions, is itself a concrete finding (§4.2).

3 Experimental Setup

The dataset (Alam et al., 2025) (config `task1a_msa`) provides four MSA splits: `train` (3,000, labeled), `dev` (500, labeled), `devtest` (500, unlabeled), and `test` (1,000, unlabeled, blind ranking). We target the MSA track exclusively. Critically, `dev`/`devtest` audio is voice-cloned while `test` audio is human-recorded — a recording-condition shift the organizers flag explicitly and that we return to in §4.3.

ASR uses Whisper medium (primary) and large-v3 (repair escalation only), both via `faster-whisper`/CTranslate2. Parser and selector are served through Ollama: `qwen2.5vl:7b` for parsing throughout, and `qwen2.5vl:7b` or `qwen3-vl:8b` for selection (§2.5).

We report accuracy, balanced accuracy, and macro F1. The gap matters here: a constant-prediction baseline (always index 0) beats random guessing on accuracy (35.8% vs. 33.4%) because label 0 is marginally over-represented, yet its macro F1 (17.6%) is roughly half of random's (33.2%), since it never predicts the other classes. Every experiment changes exactly one component relative to the previous stable configuration and is evaluated at full scale before being adopted or discarded.

4 Results

Table 1 summarizes every scored configuration. On `dev`, the production pipeline (repair enabled, Qwen3-VL selector) reaches 79.0% accuracy / 78.75% balanced accuracy / 78.80% macro F1 (395/500), roughly double the trivial baselines and well above the 39.8% official baseline.

4.1 Interpretation

The single largest dev-phase gain (+4.2 points, 74.80%→79.00%) comes from replacing only the selector model (Qwen2.5-VL→Qwen3-VL) with ASR, parsing, and repair held fixed: the bottleneck was in visual/comparative reasoning, not in recovering question structure from audio. Only the selector swap and the repair-vs.-no-repair transition were isolated at full scale under stable caching (§4.2), so we do not claim every upstream component independently improved the score. Chain-of-thought (+0.2) and few-shot retrieval (+0.0) were evaluated at full scale but did not clear the noise floor described below, and both remain off by default: negative results reported with the same rigor as our positive one. The apparent regression from an 82.00% no-repair run to the 74.80% repair-enabled baseline should *not* be read as "repair hurt": that configuration was not stable across repeated runs of the identical key.

With $n = 500$, a naive two-proportion comparison of that gap gives a combined standard error of roughly ± 2.6 points: suggestive but not conclusive, and the wrong test for a paired comparison. McNemar's test on per-record disagreements needs prediction files we did not retain (see Limitations).

Configuration	Accuracy	Bal. Acc.	Macro F1
Official baseline (Qwen2.5-Omni, MSA)	39.80%	—	—
random	33.40%	33.22%	33.20%
constant-0	35.80%	33.33%	17.57%
Full pipeline, no repair (best / worst observed run)	82.00% / 71.60%	83.81% / 71.41%	82.23% / 71.43%
+ repair (M4, adopted baseline)	74.80%	74.71%	74.71%
+ chain-of-thought	75.00%	74.86%	74.85%
+ few-shot retrieval ($k = 2$)	74.80%	74.65%	74.71%
+ Qwen3-VL selector (M6, frozen production)	**79.00%**	**78.75%**	**78.80%**
M6 on blind `test` (human audio, $n = 1{,}000$)	**65.60%**	**57.45%**	**50.44%**

Table 1: Rows above the double rule: `dev` results (n=500, synthetic audio), each changing one component vs. the row(s) above. Final row: M6's blind `test` submission (human audio, n=1,000; §4.3).

4.2 A Cache-Versioning Reproducibility Issue

The identical selector configuration key produced accuracy oscillating between roughly 71.6% and 82.0% across repeated runs, with no logged variable distinguishing them. The likely cause: our parse-stage cache key did not encode which prompt revision (§2.2) produced a given cached parse, so a configuration string naming only the selector could silently mix old- and new-prompt parses. We treat the post-transition 74.80% figure as trustworthy. Lesson: a cache key must encode every input that can change a model's output, not only the stage name; any single-run delta under roughly ten points should be read with this noise floor in mind.

4.3 Blind Test-Phase Result

All results above use synthetic `dev` audio. Our single blind submission on `test` (1,000 records, human-recorded) scored 65.60% accuracy / 57.45% balanced accuracy / 50.44% macro F1 with the frozen M6 configuration: a 13.4-point accuracy drop from `dev`, and larger 21.3- and 28.4-point drops in balanced accuracy and macro F1. We attribute this primarily to the recording-condition shift (§3): ASR tuned against clean, synthetic speech should degrade against human-recorded audio.

The disproportionate F1 drop echoes the constant-0-vs.-random pattern in §3: harder acoustics plausibly push more records into ASR/parse failures that fall through to the selector's fixed fallback index, inflating accuracy relative to F1 without the reasoning stage necessarily worsening. We report this as a plausible mechanism, not a confirmed cause.

5 Discussion

What worked. Independently cacheable stages let us attribute the largest dev-phase gain to one isolated change (§4.1), which a monolithic model could not support, and made the test-phase drop (§4.3) interpretable rather than a mystery number. Schema-constrained decoding appears to have eliminated parse-format failures entirely: every failure we observed was semantic (§2.3), never malformed.

What we would do differently. The reproducibility gap (§4.2) is a real failure mode whose likely cause is understood but not independently confirmed, and every dev-phase comparison is a single run. With hindsight, validating against even a small human-recorded sample before our one blind submission would have been the highest-value addition here. Serving all models locally via Ollama means no per-call cost and no data leaving the machine, at the cost of being bounded by an 8B model; every result is on MSA, so we scope generalization to "the architecture supports this" rather than demonstrated fact.

6 Conclusion

Decomposing spoken VQA into independently cacheable perception, language-understanding, and reasoning stages let us isolate individual design choices: upgrading the selector alone recovered more accuracy than any upstream repair or retrieval work (Table 1), and let us reason about, rather than merely observe, the drop on the blind human-recorded `test` split (§4.3). We report our two negative results, the cache-versioning bug, and that recording-condition gap with the same care as our positive result, and believe all three generalize beyond this submission.

Limitations

Every dev-phase comparison is a single run, with a ten-point swing observed under a nominally iden-

tical configuration (§4.2); our null results are "not distinguishable from no effect," and that cache anomaly is explained but not confirmed fixed. We made one blind `test` submission, so 65.6% carries no variance estimate, and our account of its F1 drop is a plausible mechanism, not a confirmed cause, absent per-record predictions — which we also lacked for the McNemar test (§4.1) and a full error analysis. Two of three ASR backends are unit-tested but never run live.

The rule-based checker (§2.3) targets the failure patterns seen in development and is not exhaustive; a degenerate parse outside them passes undetected. If ASR loses an answer choice entirely, no downstream stage can reconstruct it, and large-v3 escalation raises quality on average, not on every record. The fixed three-option schema cannot represent "fewer than three options were spoken." Some samples are intrinsically ambiguous (Appendix A). The pipeline does not yet use acoustic cues (pause locations, word-level timestamps) for boundary detection; we leave this to future work.

Ethics Statement

All models are served locally, so no audio or image data leaves the machine performing inference. We use only the dataset and splits released by the organizers, and are not aware of ethical considerations beyond those addressed by their own data release process.

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A Task Characteristics

Beyond the missing option boundaries noted in §1, three characteristics of a hand-inspected sample shaped our design. **Reasoning often exceeds object recognition**: questions may concern an activity, require reading in-image text, or draw on knowledge not visible in the image, motivating a capable VLM selector (§2.5). **Candidate answers inform reasoning**, since the options reveal which attribute matters. Finally, **some samples are ambiguous** even for a human annotator.